Pokémon Price Tracker — Project Summary

Goal

The project is a Pokémon card pricing app inspired by Collectr, with a stronger focus on **transparent and trustworthy pricing**.

The main idea:

1. Search for any Pokémon card.
2. Show current TCGplayer pricing data
3. Show recent eBay sold listings.
4. Separate those sales into meaningful categories such as raw, PSA 10, PSA 9, BGS 9.5, etc
5. Let the user decide on what is fair, using ACCURATE data from ebay last sold and TCGPlayer

The longer-term goal is to use statistics and AI to produce better price estimates while still showing users where those estimates came from. For example, I'd imagine AI would be used to classify if a card on eBay is NM, LP, etc, and also more simply to check for a graded cards grade (assuming simple NLP or something can't find this).

Current Architecture

Card Data and TCGplayer Pricing

We arbitrarily used the Pokémon TCG API first, but we were missing prices on a lot of modern cards, thus we tested other APIs. At this point we FIRSTLY just wanted to establish we could derive accurate TCGPlayer market prices from an API.

We compared Pokémon TCG API, TCGCSV, and TCGdex.

Decision: use **TCGdex as the primary source** because it had better coverage, reliability, search, and pricing freshness.

However, TCGdex has holes. For example, Lost Origin Trainer Gallery cards were missing pricing and images.

So we added **TCGCSV as a fallback:**

```
User searches card
       ↓
    TCGdex
       ↓
Card + pricing available?
    /      \
  yes      no
   ↓        ↓
 use it    TCGCSV
            ↓
       fill missing data
```

The frontend receives one normalized card object, so it does not care which source supplied the data.

We also made an important pricing rule:

> Never substitute TCGplayer `mid` or listing prices and call them "market price."

If market price is unavailable, we say so and show low/mid/high listings separately. Again, we want full transparency. ~~Fuck you collectr.~~

Search and Card Pages

Search now supports normal and alphanumeric card numbers such as:

- `199/165`
- `TG17/TG30`
- `GG69/GG70`

Search results are clickable and open a dedicated:

/card/[id]

page.

The card detail page is where we now show:

- card information

- TCGplayer pricing
- recent eBay sales

It will later also contain price-history charts and analytics. W Claude

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eBay Sold Listings

Because eBay does not provide an easily accessible public sold-history API for our use case, we currently retrieve recent sold listings through **Apify**. Apify seems to work well in providing these for us.

The app takes the exact selected card and generates an eBay search query.

Exact card
↓
Apify / eBay sold search
↓
Raw sold listings

These listings are intentionally not averaged immediately because searches contain a lot of bad comps:

- PSA 10
- PSA 9
- raw cards
- Japanese cards
- wrong card numbers
- reprints
- mystery listings
- accessories
- novelty products

Due to this, we need a classifier to seperate

Listing Classifier

We first build a **deterministic text classifier** rather than immediately using AI for the sake of computation and limits and its just simpler.

It recognizes categories such as:

RAW_NM
RAW_LP
RAW_UNKNOWN

PSA_10
PSA_9

BGS_10
BGS_9_5

CGC_10

OTHER_GRADED (e.g., psa 6)
IRRELEVANT (e.g., a clickbait raffle or some sht)

It also detects:

- grading company and grade
- raw condition when explicitly stated
- language
- wrong card numbers
- wrong sets/reprints
- mystery/raffle listings
- proxies/custom/metal cards
- accessories and other non-card products

A major principle is:

> **Do not use price to determine whether a listing is relevant.**

A $20 listing might be irrelevant because it is a raffle, while a $5,000 listing may be a legitimate PSA 10 sale.

Relevance and statistical outlier detection are separate problems.

I guess we would probably want a drop-down, like we display the price our algorithm determines its fair (e.g., ML to determine accurate pricing, with weights on date last sold and the price, then WE SHOW THE PRICES ITSELF SO THE USER CAN DETERMINE WHETHER THEY AGREE OR NOT).

Classifier Evaluation

Claude stress-tested the classifier on:

12 cards / 240 real eBay sold listings

covering modern, vintage, raw, graded, Japanese, English, reprints, and low-liquidity cards.

After improving the rules:

Clear classified: 64 (26.7%)
Rejected: 72 (30.0%)
Ambiguous: 104 (43.3%)

Confirmed false positives: 0
Confirmed false negatives: 0

The high ambiguity rate is mainly because sellers often **do not state raw card condition**.

About 78 listings had no explicit NM/LP/MP/etc.

Those correctly remain:

RAW_UNKNOWN

We decided not to guess condition from terms like:

- "Fresh Pull"
- "SIR"
- "Mint"
- price

Missing information should remain missing.

Although highly likely mint implies NM for example, we want to be extra sure.

Moving forward probably a threshold for anomalies, e.g., a card accidentally sold for 9$ when market is 90$ should be ignored.

Where AI / LangChain Comes Next

Of course the deterministic ruling isn't always gonna find what we need, and so we use LangChain to larpmax (joking)

This is now the general structure:

```
eBay listing
    ↓
Rules
    ↓
Can rules understand it?
  /        \
 yes        no
 ↓          ↓
use result   Is information missing?
               /        \
             yes         no
             ↓           ↓
          UNKNOWN     LLM fallback
```

AI should only handle genuinely semantic cases, such as:

- unclear exact card
- unusual wording
- implicit language
- unclear set/reprint relationships

It should **not** invent raw condition when the title contains no condition evidence.

We plan to use **LangChain** for this LLM fallback. LangChain will give us structured output and a clean framework around the model rather than having the app parse arbitrary LLM text.

Later, LangSmith can be used to evaluate and trace the model's decisions.

Next Development Phases

Current working foundation:

- TCGdex primary card/pricing API
- TCGCSV fallback
- card search
- card detail pages
- TCGplayer pricing
- Apify eBay sold listings
- deterministic comp classifier
- language/set/grade/condition detection

- classifier evaluation

Next:

1. LangChain semantic fallback
2. Evaluate rules + LLM
3. Statistical outlier detection
4. Grade-specific/raw price estimates
5. TCGplayer + eBay price-history charts
6. Portfolio / collection features

The overall philosophy remains:

> **Provide a simple and transparent system to INFORM vendors and buyers as ACCURATELY as possible.**